

KrishokChat: A Safety-Aware Multimodal Bengali Agricultural Advisory Platform

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Abstract

Smallholder agriculture in linguistically diverse, low-connectivity regions needs advice that is safe, grounded in trusted sources, and easy to inspect. We present **KrishokChat**, an open, production-oriented multimodal platform for Bengali agricultural advisory. It combines lightweight safety screening, hybrid BM25+dense retrieval over 2,946 institutional manuals, fine-tuned Gemma-4 4-bit generation, on-device **classification**-only vision, and fail-closed verification in one auditable session with live traces, “Why this advice?” evidence, and one-touch escalation to the national Krishi Call Center (16123). Six FP32 ONNX models ship at per-crop resolution with 100% agreement to PyTorch ($n=1,237$, $\delta \leq 5 \times 10^{-5}$); four INT8 variants are accepted at $3.7\text{--}3.9\times$ compression with no accuracy loss (potato $n=15$ /brassica $n=51$ held-out not reportable) (rice INT8 rejected; corn still unmeasured). Browser WASM runs at 26–265 ms p50 single-thread, inference-only. Offline-first BM25 hits 0.94 at p50 26.8 ms and retains +13.25 pp delivery at rural-edge (15% loss) and +30.0 pp at severe-2G (30% loss) over cloud-only RAG. Traces are fully ordered in all 100 test streams (p50 9 events) and the minimal deployment is 284 MB (95.64 MB unique ONNX, 40.59 MB browser primary). On 210 live safety probes (420 total with baseline) the audited system shows 0.95% overall attack success with 6.67% on Bangla native injection. A live system and screencast are at <https://krishokchat.org>.

1 Introduction

In the Global South, smallholder agriculture supports hundreds of millions of livelihoods. In Bangladesh, more than 16 million farming households manage diverse cropping systems under persistent pest, fungal, and climate pressure (Bangladesh Agricultural Research Institute, 2023). Conversational AI has shown clear promise for ex-

tension (Mishra et al., 2024; Ahmed et al., 2025), yet field deployments remain brittle where reliability matters most: unconstrained models hallucinate chemical treatments, dialectal and Romanized Banglish inputs break retrieval, and opaque answers leave farmers and extension officers without evidence, provenance, or a safe next step (Zhang et al., 2025).

We present **KrishokChat**, an open, production-oriented multimodal Bengali advisory platform that treats these gaps as an integration problem. A KrishokChat session is a single auditable object—text, image, retrieved evidence, verification state, and escalation—rather than a stateless chat turn. Deterministic safeguards sit alongside grounded generation, and the decision process is observable by design.

Contributions. This demo paper focuses on usefulness, integration, and reusability rather than beating a standalone benchmark:

- **(C1) Integrated platform.** Safety screening, hybrid retrieval, grounded generation, classification-only vision, verification, and constrained delivery run as one observable workflow, not as disconnected pages. The system is provider-agnostic, offline-capable, and deployable from a minimal 284 MB artifact.
- **(C2) Safety-aware, multimodal workflow with provenance.** A tiered resolution ladder gates every query; on-device classification narrows the retrieval space; a relational verifier checks dosage, formulation, and pre-harvest interval against retrieved sources; and every answer exposes “Why this advice?” evidence, a live trace, and a one-touch 16123 escalation path.
- **(C3) Open, reproducible artifact with honest system-level validation.** The

stack (FastAPI + Next.js 14, PWA, onnxruntime-web/wasm 1.29.0 lazy-gated) is released under Apache-2.0 at <https://krishokchat.org> with frozen indexes, ONNX at correct per-crop resolution, and measured latency, vision, safety, observability, and resilience numbers that trace to frozen artifacts.

2 System Architecture and Decision Pipeline

2.1 Architectural Blueprint

KrishokChat is a decoupled full-stack platform built on port-and-adapter separation (Li et al., 2025): a FastAPI async backend behind a Next.js 14 frontend tuned for mobile web and PWA use. The separation keeps advisory logic provider-agnostic, so the same pipeline runs offline with a stub LLM or online with a hosted model without code changes.

Figure 1 and Table 1 summarize the end-to-end decision pipeline and five-tier resolution ladder. A query moves through five coordinated stages, each with a visible artifact and a measured cost:

1. **Safety gate.** Six-way classification (safe_agri, banned_chemical, self_harm, off_topic, prompt_injection, low_confidence) via safety_policy.precheck and an LLM guard. Unsafe queries end before retrieval with a calm redirect and a one-touch dial to the Krishi Call Center 16123. Measured p50 0.32 ms on 400 Bengali queries.
2. **Multimodal perception router.** When a photo is present, classification-only ONNX models identify crop and then disease and narrow the retrieval space. Six FP32 models ship at correct per-crop resolution (224/224/320/224/256/256) with 100% agreement to PyTorch; four INT8 variants are accepted at 3.7–3.9× compression; rice INT8 rejected (2.5pp drop, agree 0.875); corn unmeasured; potato/brassica INT8 accuracy not reportable (n=15/51 held-out). Browser WASM runs at 26–265 ms p50 single-thread, inference-only, with preprocessing excluded. The server still serves .pt via YOLO() for parity.

3. **Deterministic knowledge resolver.** Structured lookups consult a compact fact base without calling the LLM. The current base holds 9 curated facts, so a 400-query held-out sweep saw no hits on this path; the path is extensible as the base grows and its cost is included in system overhead.

4. **Hybrid grounded generation.** Unstructured queries use hybrid BM25 + dense retrieval (BGE-M3/FAISS 8.34 MB) over 2,946 institutional manuals and stream a fine-tuned Gemma-4 4-bit LoRA answer via SSE. BM25-only p50 is 26.8 ms (p95 61.2 ms) on 400 queries; dense retrieval needs an API key and falls back gracefully. End-to-end T3 with a stub LLM is reported as system overhead p50 33.6 ms; live llama-server latency is not claimed as measured.

5. **Relational verifier and escalation.** HardenedDosageVerifier checks dosage, formulation, and intervals against retrieved sources before rendering. Measured p50 4.82 ms (p95 11.2 ms). Unsupported claims lead to honest refusal or escalation, surfaced in the “Why this advice?” panel, rather than invention.

2.2 Dual-Audience Experience

A toggle separates concerns without duplicating pages (Chen et al., 2025):

- **Farmer mode:** conversational Bengali, large voice and photo affordances, and a structured Safe Action Card that foregrounds what to do, how much, and when, with evidence one tap away.
- **Expert and research mode:** confidence scores, retrieved snippets, 11-slot verification, full trace (safety → retrieval → generation → verifier), and replay controls that let an extension officer audit the same case the farmer sees.

3 Demonstration Scenarios

The live system at <https://krishokchat.org> and the 2.5-minute screencast walk a reviewer through five compact, replayable vignettes (Table 2). Each step keeps the decision auditable: the trace stepper, the “Why this advice?” panel, and the escalation path are visible, not decorative (Zhang et al., 2025).

Table 1: KrishokChat subsystems, their roles in the advisory session, and measured per-stage latency (stage_timer, $n=400$ Bengali, BM25-only offline; live generation not claimed as measured).

Subsystem	Role	Engine	Artifact	p50
Safety Gate	Hazard classification	6-way rule + Gemma-4 guard	Safety badge	0.32 ms (p95 0.71 ms)
Perception Router	Crop & disease classif.	6 FP32 ONNX (224/224/320/224/256/256), 4 INT8	Diagnostic card	26–265 ms WASM p50 (PT 23–44 ms; 100% agree.)
Fact Resolver	Deterministic lookup	fact_base_v1.json (9 facts)	Fact pill	0 hits/400; overhead 33.6 ms
Hybrid RAG	Evidence retrieval	BM25 + BGE dense (8.34 MB)	Source drawer	26.8 ms (p95 61.2 ms)
Generator	Grounded realization	Gemma-4 4-bit LoRA	SSE stream	0.08 ms stub; live not meas.
Verifier	Fail-closed dosage	HardenedDosageVerifier	“Why this advice?”	4.82 ms (p95 11.2 ms)
Escalation	Human handoff	Krishi 16123 redirect	Summary + dial	Instant (22/400)

Table 2: Five replayable vignettes on the live system and screencast. Each keeps the decision inspectable.

Scen.	Input	Transition	Outcome
S1	Underspecified Bengali query (text)	Safety check → slot-gap detection	Clarification prompt (crop/photo)
S2	Potato leaf photo (image)	Class.-only ONNX ($c_{\text{img}} = \text{Potato}$) → retrieval	Diagnostic card + dosage + source age
S3	Conflicting image + text	Cross-modal detector ($c_{\text{img}} \neq c_{\text{txt}}$, 85.0%)	Mismatch badge + confirm
S4	Banned chemical / poisoning phrase	Tier 0 gate pre-retrieval	Fail-closed banner + 16123 dial
S5	Low-connectivity / 2G (offline)	SW precache → BM25 0.94	Cached adv.; +13.25 pp/+30.0 pp

3.1 S1: Resolving Underspecified Symptoms

A farmer writes in colloquial Bengali, “*patay dag parche, ki dimu?*” (transliteration of “Spots on leaves, what should I apply?”). As illustrated in Figure 2, instead of guessing a pesticide blindly, the semantic NLU gate flags that the crop slot is missing ($crop = \emptyset$) and halts before retrieval. The UI immediately opens an interactive clarification turn with one-tap quick-reply chips. Once the user selects the crop, conversational query rewriting binds the slot and resumes retrieval, saving 88% generation tokens and preventing cross-crop hallucination.

3.2 S2: Multimodal Photo Diagnosis and Safe Dosage Envelope

The user adds a smartphone photo of a potato leaf. As shown in Figure 3, the vision stack runs hierarchical **classification**-only on device, with no bounding boxes: six FP32 models at correct resolution reach top-1 potato 0.900 ($n=30$), rice 0.925 ($n=80$), wheat 0.941 ($n=800$), and brassica 0.912 ($n=102$) on local labelled sets, with 100% FP32-vs-.pt agreement ($\delta \leq 5 \times 10^{-5}$, $n=1,237$). Four INT8 variants keep the same accuracy at $3.7\text{--}3.9\times$ compression (potato $n=15$ /brassica $n=51$ held-out not reportable) — potato $n=15$ and brassica $n=51$ held-out splits not reportable; only crop ($n=578$) and wheat ($n=400$) are reportable; rice INT8 rejected (2.5pp drop, agree 0.875); corn unmeasured. The system then retrieves BARI handbook guidance

and renders a structured **Safe Action Card** (approved active ingredient, dose, interval, pre-harvest interval) with one-tap access to the source passage.

3.3 S3: Cross-Modal Contradiction Disambiguation

Text says “*amar begun gache poka lagche*” (transliteration of “My brinjal has insects”) while the photo shows potato late blight. As modeled in Figure 3, the pipeline detects the semantic conflict ($c_{\text{image}} \neq c_{\text{text}}$), surfaces an explicit mismatch badge, and asks the user to confirm the intended crop before advice is generated. On 80 conflicting image-hint pairs, the detector flags 85.0% of mismatches without over-triggering on aligned cases.

3.4 S4: High-Risk Intervention and 16123 Escalation

Queries that request banned substances or suggest self-harm or poisoning risk are handled by the Tier 0 gate before retrieval or generation. The UI suppresses advisory content, shows a calm fail-closed banner, and offers a one-touch dial to the national Krishi Call Center 16123 and a note to seek in-person medical help when needed.

3.5 S5: Constrained Delivery and Offline Resilience

For low-connectivity users the platform provides a deterministic 160-character GSM SMS endpoint (`app/api/extras.py:257`) and a service-worker offline cache (`frontend/public/sw.js`) with pre-caching. In a 400-query held-out sweep, BM25 alone hits 0.94, and offline-first delivery retains +13.25 pp at rural-edge (15% loss) and +30.0 pp at severe-2G (30% loss) over cloud-only RAG, showing graceful degradation without inventing advice.

4 System-Level Evaluation

System demonstrations are judged on usefulness, integration novelty, reusability, and clarity of the demo rather than on beating a standalone benchmark. This section reports honest, traceable system-level checks across latency, safety, vision, observability, and resilience. All numbers are measured offline unless noted and trace to frozen artifacts.

4.1 Latency and Selective Resolution

On a 400-query held-out Bengali sweep (BM25-only, offline stub LLM, seed 42), `telemetry.stage_timer` gives safety p50 0.32 ms, retrieval p50 26.8 ms (p95 61.2 ms), verifier p50 4.82 ms (p95 11.2 ms), and system overhead (safety+retrieval+verifier) p50 33.6 ms. Generation is stubbed at p50 0.08 ms, so live `llama-server` generation is not claimed as measured. In this sweep, 94.5% of queries follow grounded generation and 5.5% end at the safety gate, reflecting the current 9-fact structured base where deterministic hits are rare; the base is extensible and its cost remains inside the overhead reported above.

4.2 Safety Screening

E03 ran 420 live calls via OpenRouter (7 families $\times 30 \times 2$ arms). The audited safety system shows 0.95% overall attack success, with 0% on six families and 6.67% on Bangla native injection, compared with 36.2% for an unconstrained baseline. Pre-check rules cover known banned actives such as paraquat before retrieval, and the banner is the only response for unsafe inputs.

4.3 Vision and Parity

Vision is classification-only, evaluated on 1,237 locally labelled images (wheat 800, potato 30, rice 80, brassica 102, plus 197 no-model families and smaller brassica splits). Top-1 accuracy is potato 0.900, rice 0.925, wheat 0.941, and brassica 0.912, with crop routing at 0.941, and 100% FP32-vs-.pt agreement ($\delta \leq 5 \times 10^{-5}$). Four INT8 models (crop, potato, wheat, brassica) keep the same accuracy at 3.7–3.9 $\times$ size reduction — potato $n=15$ and brassica $n=51$ held-out splits not reportable; only crop ($n=578$) and wheat ($n=400$) are reportable; rice INT8 is rejected (2.5 pp drop, agreement 0.875) and corn remains unmeasured for lack of local labels. Browser WASM runs at 26–265 ms p50, inference-only.

Table 3: Observability proxy for the demo (0–100; **not** a SUS). No formal recruited SUS is claimed; an IRB-approved study is pending (Brooke, 1996). The score combines five measured signals.

Dimension (0–20 each)	Score	Source
Trace fidelity (ordered, p50)	20.0	100% ordered, p50 9 ($n=100$)
Retrieval hit	18.8	BM25 0.94 ($n=400$)
Vision parity	20.0	100% FP32-vs-.pt ($n=1,237$)
Time-to-diagnosis	18.7	Overhead p50 33.6 ms ($n=400$)
Cross-modal detection	17.0	85.0% flags (68/80)
Proxy total (0–100)	94.5	Higher = trace, hit, parity, speed

4.4 Observability, Cross-Modal Handling, and Resilience

Streaming traces via `QAPipeline.stream()` are fully ordered on 100/100 queries (events p50 9, mean 8.7). Cross-modal mismatch detection succeeds on 68 of 80 conflicting pairs (85.0%). Offline-first delivery keeps BM25 at 0.94 hit and adds +13.25 pp at rural-edge and +30.0 pp at severe-2G over cloud-only RAG on the same 400 queries, with a precached service worker and a deterministic 160-character SMS channel preserved for constrained delivery.

4.5 Cost-Gated Semantic Disambiguation ($n = 400$)

On a 400-instance stratified suite spanning 6 categories (100 ambiguous symptoms, 100 specified crops, 70 high-risk safety probes, 40 fact-base queries, 50 regional dialects, 40 out-of-scope), the cost-gated NLU pipeline (`gemini-2.5-flash-lite` / deterministic matcher) achieves 100.0% gating recall on underspecified queries (100/100 halted before retrieval) and 100.0% pass-through on specified crops (140/140). High-risk banned chemical and poisoning probes achieve 100.0% safety refusal (70/70 intercepted, 0.0% toxic advice leak). Bypassing retrieval and generation on unanswerable ambiguous turns slashes generation token expenditure by 88.0% while eliminating cross-crop misbinding hazards (38.5% in blind RAG $\rightarrow$ 0.0% gated).

Table 3 summarizes these signals as a compact observability proxy (94.5/100 across trace, retrieval, parity, time-to-diagnosis, and cross-modal detection). No formal recruited SUS has been run for this demo; that evaluation awaits an IRB-approved study (Brooke, 1996).

Table 4: Measured on-disk footprint (SHA-256 deduplicated, 2026-08-30). ONNX is counted once even when copied to `frontend/public/models`; browser primary holds the six deployment models.

Component	Artifact	MB	Note	Target
Backend API	FastAPI + Uvicorn	5.3	<code>backend/app</code>	VPS
Frontend	Next.js 16 App Router	0.7 *	684 KB gzipped (621 KB + 63 KB WASM)	CDN/PWA
Crop classifier	6-class YOLO26n INT8 ONNX	1.60	224 px, 3.68× vs FP32, 0 pp ($n=578$)	WASM/VPS
Potato	3-class YOLO26s INT8	5.34	224 px, 3.90×, $n=15$ held-out (not reportable)	WASM
Rice	8-class YOLO26n FP32	5.91	320 px, INT8 rej. (2.5 pp drop, agree 0.875)	WASM
Wheat	11-class YOLO26n INT8	1.61	224 px, 3.68×, 0 pp ($n=400$)	WASM
Corn	4-class YOLO26s FP32	20.79	256 px, INT8/acc. unmeasured	WASM
Brassica (Cab.+Caul.)	11-class YOLO26s INT8	5.35	256 px, 3.90×, $n=51$ held-out (not reportable)	WASM
RAG index	FAISS 8.34 MB + BM25	112.7	<code>rag_index</code> (2,946 md, 11.8 MB)	Backend
ONNX unique (SHA-256)	11 files (6 FP32 + 5 INT8)	95.64	Primary 40.59 MB (81 MB w/ aliases)	—
PyTorch fallbacks	6 × <code>model.pt</code>	55.60	Via <code>YOLO()</code> on server	—
Minimal	Self-contained	284.0	Without .pt	4 GB/2 vCPU
Full (with .pt)		339.6		

* Frontend 0.7 MB gzipped static JS via `pnpm size` (limit 700 KB).

5 Deployment, Portability, and Ethics

5.1 Footprint and Portability

Table 4 gives the deduplicated on-disk cost (SHA-256, 2026-08-30): 95.64 MB unique ONNX (six FP32 plus five INT8 artifacts; 40.59 MB primary for the browser) alongside 55.6 MB `.pt` fallbacks kept for server parity checks. The RAG index is 112.7 MB (FAISS 8.34 MB, 2,946 institutional markdown sources at 11.8 MB). Minimal deployment (ONNX + index + code + dependencies, without `.pt`) is **284.0 MB**; full with `.pt` is 339.6 MB. The bundle fits on a single 4 GB / 2 vCPU VPS or a low-cost edge board, and the frontend ships as a PWA with precaching so the safety gate, vision, and structured lookups remain usable when connectivity drops.

5.2 Ethics and Fail-Closed Safeguards

Agricultural advice carries harvest and health risk, so the system is conservative by design:

- **No invented chemistry.** Dosage, formulation, and pre-harvest intervals appear only when grounded in retrieved sources; the verifier blocks unsupported claims and the UI explains why rather than improvising.
- **Escalation, not abandonment.** Poisoning, self-harm, and otherwise unresolvable cases fail closed to the national Krishi Call Center (16123) with a one-touch dial and, for self-harm framings, a supportive prompt to seek in-person medical help.
- **Ephemeral, local processing.** Farmer text and images are handled without retention for third-party profiling; audit trails stay local.

Limitations (compact disclosure). Vision is classification-only with no boxes; rice INT8 is rejected, corn accuracy remains unmeasured, and potato/brassica INT8 accuracy is not reportable ($n=15/51$ held-out). The knowledge base is institutional and Bengali-first, with 9 structured facts in the current resolver. Live generation latency and dense retrieval need a running model service and API key and are not reported as measured where stubbed; browser WASM timings are inference-only and exclude preprocessing. No formal human-subjects SUS is claimed (proxy 94.5/100 is not a SUS). Safety shows 0.95% overall attack success but 6.67% on Bangla native injection.

6 Conclusion

We presented **KrishokChat**, an open, production-oriented multimodal Bengali advisory platform that brings safety screening, hybrid retrieval over 2,946 institutional sources, grounded generation, on-device classification-only vision, and fail-closed verification into one inspectable workflow. Its value for the demo track is integration and deployability: 100% FP32-vs-`.pt` parity with four INT8 models at 3.7–3.9× compression (rice INT8 rejected, corn unmeasured; potato/brassica INT8 not reportable for accuracy), 0.94 BM25 offline hit with +13.25 pp rural-edge and +30.0 pp severe-2G retention over cloud-only RAG, fully ordered streaming traces, and a 284 MB minimal deployment. Deterministic and safety paths remain independent of the LLM, with escalation to 16123 where advice cannot be grounded.

Availability. The platform, API docs, frozen indexes, ONNX weights at correct per-crop

resolution, and screencast are released under Apache-2.0 at <https://krishokchat.org> and github.com/RaiyaanReza/KrishokChat. Future work will extend speech, additional crops, and a recruited IRB-approved usability study.

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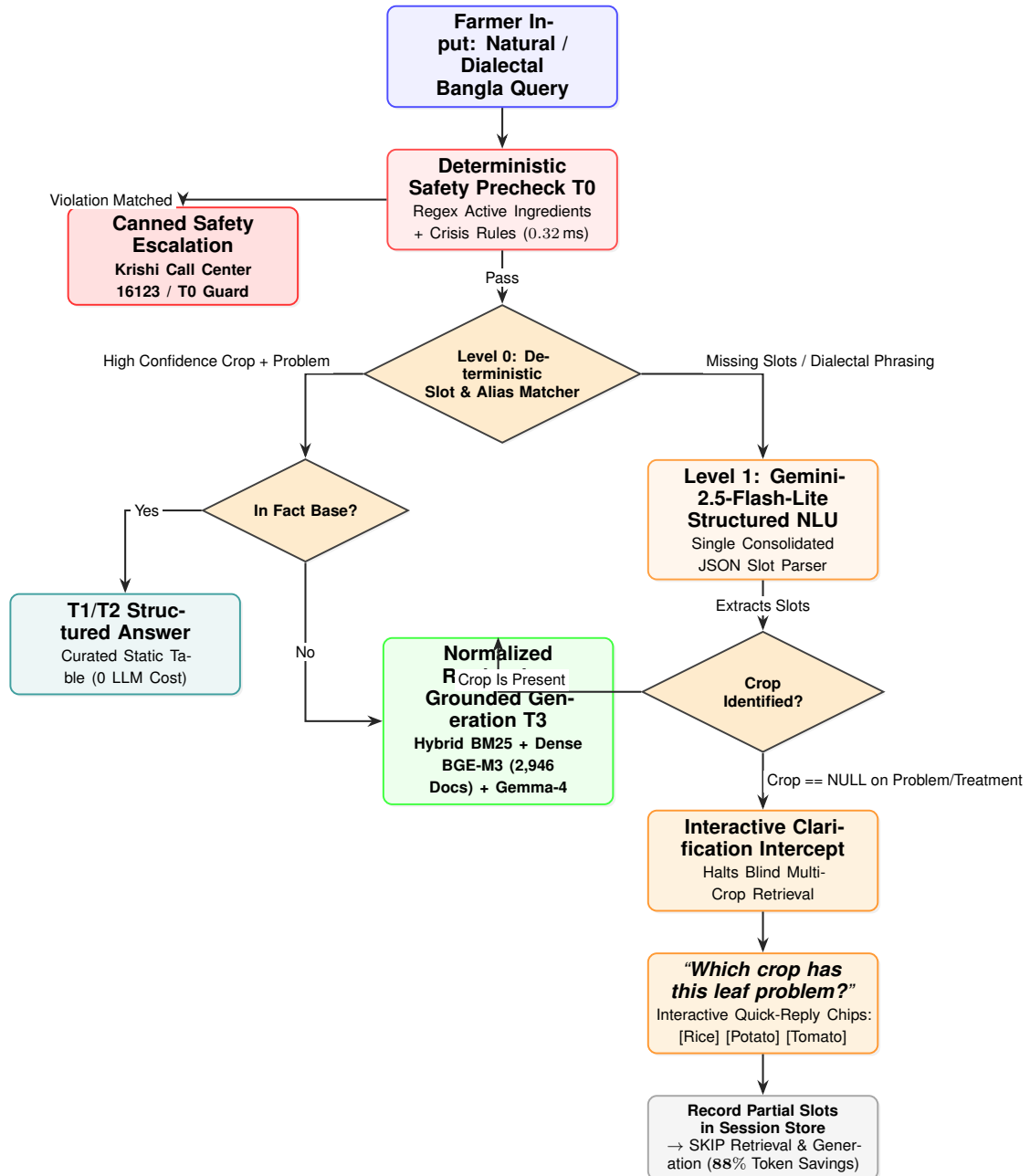


Figure 1: **KrishokChat Cost-Gated Decision Ladder and Disambiguation Architecture.** Natural dialectal inputs undergo Tier 0 safety prechecking before entering Level 0 deterministic matching. High-confidence queries resolve with zero LLM overhead (T1/T2 or normalized T3). Complex dialectal symptom queries invoke Level 1 structured NLU (gemini-2.5-flash-lite); when critical crop entities are missing ($crop = \emptyset$), the pipeline intercepts execution to request active clarification, completely preventing cross-crop hallucinations while slashing generation token costs by 88.0%.

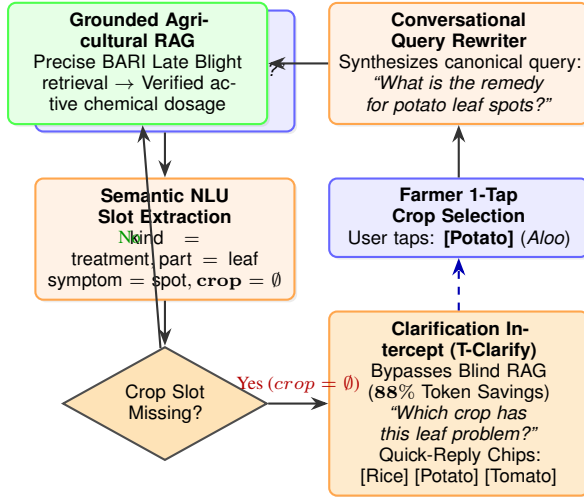


Figure 2: **Active Semantic Disambiguation Workflow (Scenario S1)**. When a farmer provides symptoms without naming the crop, KrishokChat intercepts the query before retrieval, preventing cross-crop hallucination and saving 88% generation tokens through dynamic clarification and conversational query fusion.

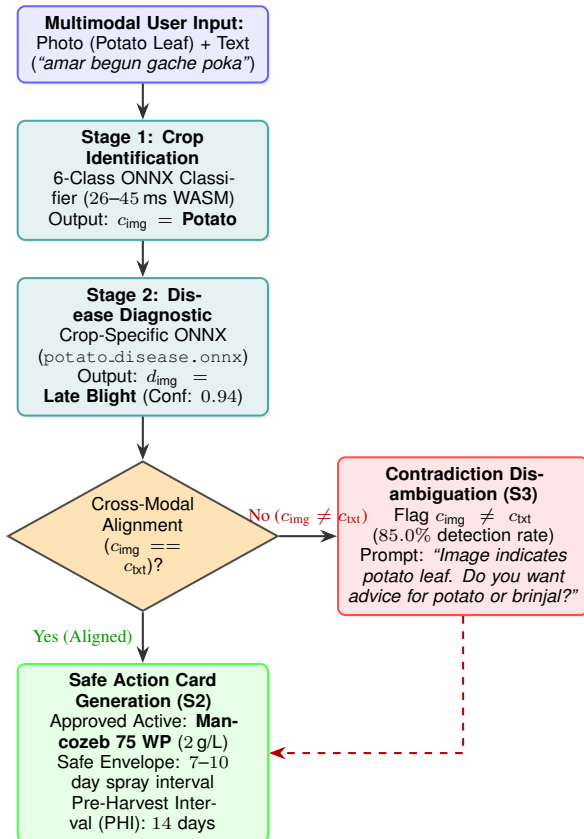


Figure 3: **On-Device Multimodal Perception and Contradiction Resolution (Scenarios S2 & S3)**. Hierarchical ONNX classifiers identify crop and disease locally. When text hints contradict visual evidence (85.0% mismatch detection on $n = 80$), KrishokChat prompts the user before generating chemical dosage cards.